

Evidence-Gap Assessment: Technical Engineering Practices

Base Practice	Status	Rationale
Prepare for software system design definition	EvidenceStatus.STRONG_EVIDENCE	The claims regarding OptArrow's architecture and integration capabilities demonstrate a clear alignment with implementing a design definition strategy. Claim-1 and Claim-2 illustrate this alignment, with supportive evidence and observed verification, confirming preparation strategies and selection of design principles.
Establish designs related to each software system element	EvidenceStatus.STRONG_EVIDENCE	The transformation of architectural and design characteristics into implemented capabilities is evidenced by Claim-2 regarding performance optimizing strategies, and Claim-5 validating functional realizations. The claims provide strong evidence of establishing and refining interfaces, as well as examining design alternatives.
Assess alternatives for obtaining software system elements	EvidenceStatus.PARTIAL_EVIDENCE	While the claims view Apache Arrow as a vital component, evidencing some analysis and selection of technological alternatives, they primarily focus on realized integration rather than detailed assessments of competing alternatives. Thus, evidence only partially fulfills the assessment component.
Manage the design	EvidenceStatus.STRONG_EVIDENCE	The thorough documentation and verification of OptArrow's capabilities and key elements across claims underscore Claim-5's emphasis on procedural and traceable design management. The claims back the capturing of design rationale and establishment of traceability.